PT.SUMITOMO ELECTRIC WINTEC INDONESIA

$\overline{X}$ - R CONTROL CHART

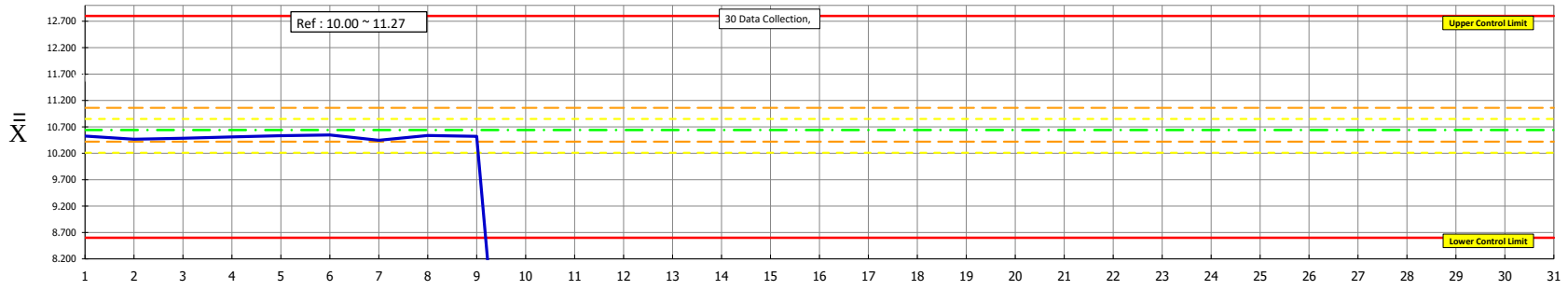
No. Dokumen	**4-QAS-PP07-001**
Revisi	**00**
Tgl Berlaku	**2-Dec-16**

MACHINE	SECTION	OPERATION	DATE CONTROL LIMITS CALC.	SPECIFICATION	COURSE NO.	CONTROL ITEM (Δ) YES / NO
SCRAPE	**QI**	**Scrape Tester**		min **8.16** max **-**		

MONTH	CHARACTERISTIC	SAMPLE SIZE / FREQUENCY	TYPE / SIZE : **D2IDW4-B-2 / 0.70**
May-25	**X Scrape**	n = **3**	PIC : **ANDI JAYA**

X = AVERAGE $\overline{X}$ = **10.640** U.C.L. = $\overline{\overline{X}} + A_2\overline{R}$ = **12.8000** L.C.L. = $\overline{\overline{X}} - A_2\overline{R}$ = **8.6000**

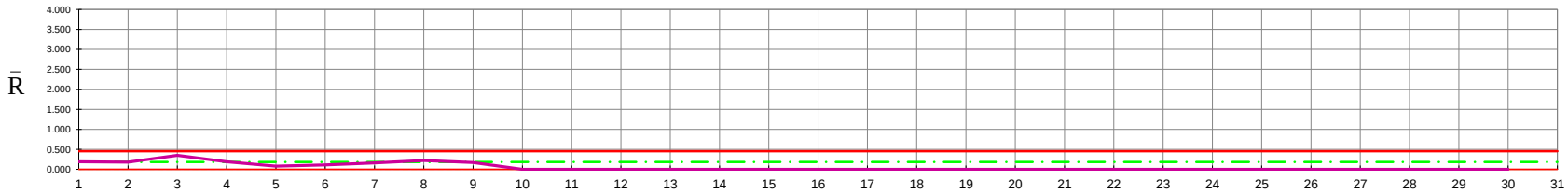
AVERAGE (X BAR CHART)



$\overline{R}$ = AVERAGE R = **0.183** U.C.L. = $D_4\overline{R}$ = **0.454** L.C.L. = $D_3\overline{R}$ = **0.00**

RANGES (R - CHART)

Special Cause Marker Legend:
Test 1 ● Test 2 ◆ Test 3 ▲ Test 4 ■



SPECIAL CAUSES

- ☑ TEST 1: ANY POINT OUT OF CONTROL LIMIT
- ☑ TEST 2: 7 POINTS IN A ROW ON SAME SIZE OF CENTER LINE
- ☑ TEST 3: 6 POINTS IN A ROW ALL INCREASING OR DECREASING
- ☑ TEST 4: 14 POINTS IN A ROW, ALTERNATING UP & DOWN
- ☑ TEST 5: 2 OUT OF 3 POINTS > 2 STDEV FROM CENTER LINE (SAME SIDE)
- ☑ TEST 6: 4 OUT OF 5 POINTS > 1 STDEV FROM CENTER LINE (SAME SIDE)
- ☑ TEST 7: 15 POINTS IN A ROW WITHIN 1 STDEV OF CENTER LINE (EITHER SIDE)
- ☑ TEST 8: 8 POINTS IN A ROW > 1 STDEV FROM CENTER LINE (EITHER SIDE)

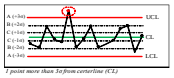
ACTION INSTRUCTION

1. _____
2. _____
3. _____
4. _____
5. _____

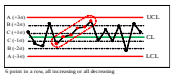
SUBGROUP SIZE	A_2	d_2	D_3	D_4
2	1.881	1.128	0	3.267
3	1.023	1.693	0	2.574
4	0.729	2.059	0	2.281
5	0.577	2.326	0	2.115
6	0.483	2.534	0	2.004
7	0.4193	2.704	0.0756	1.9244
8	0.3726	2.847	0.1361	1.8639

Special Cause

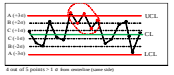
Titik berada di luar garis merah atau lebih dari 3 Ơ



6 titik berada dalam satu urutan secara terus menerus, baik posisi naik ataupun turun



4 atau 5 titik lebih dari 1Ơ dengan posisi yang sama dan titik keluar dari garis merah atau 3Ơ



Month																														
Date	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
DATA 1	10.420	10.390	10.330	10.410	10.500	10.490	10.370	10.410	10.420																						
DATA 2	10.550	10.440	10.440	10.520	10.520	10.550	10.440	10.570	10.550																						
DATA 3	10.610	10.570	10.680	10.600	10.580	10.600	10.530	10.630	10.590																						
N	3	3	3	3	3	3	3	3	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AVERAGE, X	10.527	10.467	10.483	10.510	10.533	10.547	10.447	10.537	10.520																						
Range	0.190	0.180	0.350	0.190	0.080	0.110	0.160	0.220	0.170																						
Moving Range	0.000	-0.060	0.017	0.027	0.023	0.013	-0.100	0.090	-0.017																						
SHIFT 1	ANJ	ANJ	ANJ	ANJ			ANJ	ANJ	ANJ	ANJ	ANJ			ANJ	ANJ	ANJ	ANJ	ANJ			ANJ	ANJ	ANJ	ANJ	ANJ			ANJ	ANJ	ANJ	ANJ
SHIFT 2																															

CALCULATION

$\overline{X}$ = 10.5078

$\overline{R}$ = 0.1833

s_{ST} = 0.1083 s_{LT} = 0.090

THE PROCESS MUST BE IN CONTROL BEFORE CAPABILITY CAN BE DETERMINED

$C_{PK} = \text{MIN}\left[\frac{X - LSL}{3s_{ST}} \quad \frac{USL - X}{3s_{ST}}\right]$

$= \text{MIN}\left[5.87 \quad 7.06\right]$

$= 5.87$

$Cp = \frac{USL - LSL}{6s_{ST}}$

$= 6.46$

CHECKED BY	APPROVED BY